Class Test 1

ML for EE (EEE G513)

February 26, 2026

Total Time: 75 Minutes; Total Marks: 45 (15%)
Type: Closed Book
Write all steps in derivations. Do not skip. State your assumptions clearly
Answer questions in order

Logistic Regression (10 marks)

1. Consider the data in Figure 1, where we fit the model $p(y = 1|\mathbf{x}, \mathbf{w}) = \sigma(w_0 + w_1x_1 + w_2x_2)$. Suppose we fit the model by maximum likelihood, i.e., we minimize

$$J(\mathbf{w}) = -\ell(\mathbf{w}, \mathcal{D}_{\text{train}}) \qquad \text{(3 marks)}$$

where $\ell(\mathbf{w}, \mathcal{D}_{\text{train}})$ is the log likelihood on the training set. Sketch a possible decision boundary corresponding to $\hat{\mathbf{w}}$. (Copy the figure first (a rough sketch is enough), and then superimpose your answer on your copy, since you will need multiple versions of this figure). Is your answer (decision boundary) unique? How many classification errors does your method make on the training set?

2. Now suppose we regularize only the w_0 parameter, i.e., we minimize

$$J_0(\mathbf{w}) = -\ell(\mathbf{w}, \mathcal{D}_{\text{train}}) + \lambda w_0^2 \qquad \text{(3 marks)}$$

Suppose λ is a very large number, so we regularize w_0 all the way to 0, but all other parameters are unregularized. Sketch a possible decision boundary. How many classification errors does your method make on the training set?

3. Now suppose we heavily regularize only the w_1 parameter, i.e., we minimize

$$J_1(\mathbf{w}) = -\ell(\mathbf{w}, \mathcal{D}_{\text{train}}) + \lambda w_1^2 \qquad \text{(3 marks)}$$

Sketch a possible decision boundary. How many classification errors does your method make on the training set?

4. Write the expression of the Loss function $J(\mathbf{w})$? (1 mark)

Probability (7 marks)

1. Define three random variables, X, Y and Z by the following procedure. Roll a six-sided die and a four-sided die. Now flip a coin. If the coin comes up heads, then X takes the value of the six-sided die and Y takes the value of the four-sided die. Otherwise, X takes the value of the four-sided die and Y takes the value of the six-sided die. Z always takes the value of the sum of the dice. (2 marks)

   (a) What is $P(X)$, the probability distribution of this random variable?

   (b) What is $P(X, Y)$, the joint probability distribution of these two random variables?

   (c) Are X and Y independent?

   (d) Are X and Z independent?

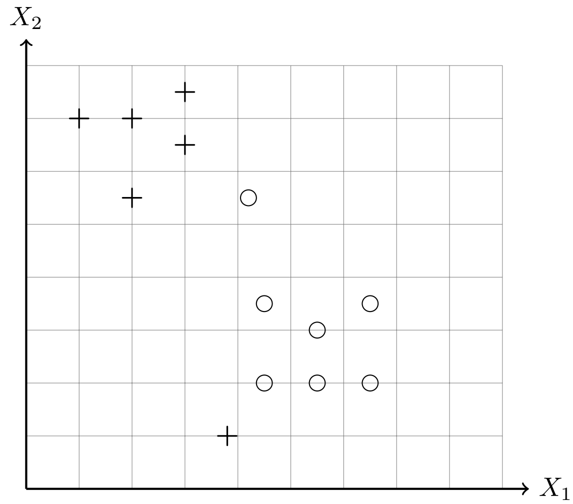

Fig. 1

2. Define two random variables X and Y by the following procedure. Flip a fair coin; if it comes up heads, then $X = 1$, otherwise $X = -1$. Now roll a six-sided die, and call the value U. We define $Y = U + X$. (2.5 marks)
   (a) What is $P(Y|X = 1)$?
   (b) What is $P(X|Y = 0)$?
   (c) What is $P(X|Y = 7)$?
   (d) What is $P(X|Y = 3)$?
   (e) Are X and Y independent?

3. Magic the Gathering is a popular card game. Cards can be land cards, or other cards. We consider a game with two players. Each player has a deck of 40 cards. Each player shuffles their deck, then deals seven cards, called their *hand*. The rest of each player's deck is called their *library*. Assume that player one has 10 land cards in their deck and player two has 20. Write L_1 for the number of lands in player one's hand and L_2 for the number of lands in player two's hand. Write L_t for the number of lands in the top 10 cards of player one's library. (2.5 marks)
   (a) Write $S = L_1 + L_2$. What is $P(\{S = 0\})$?
   (b) Write $D = L_1 - L_2$. What is $P(\{D = 0\})$?
   (c) What is the probability distribution for L_1?
   (d) Write out the probability distribution for $P(L_1|L_t = 10)$.
   (e) Write out the probability distribution $P(L_1|L_t = 5)$.

MLE (6 marks)

1. **Fitting a Binomial Model:** You encounter a deck of Martian playing cards. There are 87 cards in the deck. You cannot read Martian, and so the meaning of the cards is mysterious. However, you notice that some cards are blue, and others are yellow.
   (a) You shuffle the deck, and draw one card. It is yellow. What is the maximum likelihood estimate of the fraction of blue cards in the deck? (1 mark)
   (b) You repeat the previous exercise 10 times, replacing the card you drew each time before shuffling. You see 7 yellow and 3 blue cards in the deck. What is the maximum likelihood estimate of the fraction of blue cards in the deck? (2 marks)

(c) You repeat the previous exercise 10 times, **not** replacing the card you drew each time before shuffling. You see 7 yellow and 3 blue cards in the deck. What is the maximum likelihood estimate of the fraction of blue cards in the deck? (3 marks)

Applied Regression (4 marks)

1. Suppose we collect data for a group of students in the course EEE G404 with variables X_1 = hours studied, X_2 = undergrad CGPA, and Y = receive an A. We fit a logistic regression and produce estimated coefficient, $\hat{\beta}_0 = -6, \hat{\beta}_1 = 0.1, \hat{\beta}_2 = 0.5$. (2+2 marks)

   (a) Estimate the probability that a student who studies for 50 hours and has an undergrad CGPA of 7.9 gets an A in the class.

   (b) How many hours would the student in part (a) need to study to have a 50% chance of getting an A in the class?

Naive Bayes (10 marks)

1. Derive the Naive Bayes classifier mechanism from scratch as done in the class. Explain all the steps with proper reasoning.

KNN Code (8 marks)

1. Fill in the missing code snippets. In your answers sheets, you have to fill in just the missing TODOS. Do not write the entire code.

```python
def knn_predict(X_train, y_train, X_test, k=3):
    """
    Predicts labels for test data using the k-Nearest Neighbors algorithm.

    Parameters:
    X_train (np.array): Training features
    y_train (np.array): Training labels
    X_test (np.array): Testing features
    k (int): Number of neighbors to consider

    Returns:
    np.array: Predicted labels for the test set
    """
    predictions = []

    for test_point in X_test:
        # 1. Calculate Euclidean distances between test_point and all x in X_train
        # Hint: Use np.sqrt and np.sum for vectorization
        distances = ### TODO 1: YOUR CODE HERE ###

        # 2. Find the indices of the k smallest distances
        # Hint: np.argsort is useful here
        k_indices = ### TODO 2: YOUR CODE HERE ###

        # 3. Get the labels associated with those k indices from y_train
        k_nearest_labels = ### TODO 3: YOUR CODE HERE ###

        # 4. Perform a majority vote to find the most common label
        # Hint: Use Counter(k_nearest_labels).most_common(1)
        most_common = ### TODO 4: YOUR CODE HERE ###
        predictions.append(most_common[0][0])

    return np.array(predictions)
```

Listing 1: k-Nearest Neighbors Implementation